

RapidBoxForest: Link-Envelope-Guided Box Forests for Low-Overhead Scene Construction in Multi-Query Manipulation

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Abstract—Reusable planning structures amortize work in multi-query manipulation, but certified convex regions can be costly to construct and update. RapidBoxForest (RBF) uses axis-aligned configuration-space (C-space) boxes as reusable regions: it validates boxes with endpoint-to-link swept-capsule bounds, connects policy-accepted boxes into a scene forest and typed corridor graph, and stores kinematics-keyed envelope evidence in a Lifelong Envelope Cache Tree (LECT) separately from scene labels. The certificate model is explicit: claims apply only to conservative box unions or finite-cover oriented bounding box (OBB) corridor unions; repaired, shortened, post-processed, filter-derived, or otherwise uncovered paths are accepted only after query validation. Saved-catalog evaluations fix start-goal and obstacle instances before planning and count success only after fixed-step post-run validation. With a prebuilt depth-23 LECT cache, RBF replay on Shelf+KUKA LBR iiwa yields medians of 0.171 s reusable build time, 0.077 s per-query online time (Online/q), and 0.111 s five-query amortized time, compared with 2.770 s live hierarchical interval forward kinematics with five split levels (HIFK-5) under the same depth and validation-budget setting. In saved random-scene catalogs, the bidirectional rapidly-exploring random tree baseline (RRT-Connect) has lower median Online/q in five of six points, but higher validated path-length ratios than RBF in all six; compared with Batch Informed Trees (BIT*), RBF has lower median Online/q in all six, while BIT* has lower ratios in the four points involving Universal Robots UR5 and Franka Emika Panda. Scene-cache maintenance rows report $153.3\text{--}207.5\times$ insertion and $943.6\text{--}3354.0\times$ removal speedups over a 15-obstacle rebuild; these rows exclude snapshot, memory, and end-to-end update-and-query costs. Code: github.com/tianyu1997/RapidBoxForest.

Index Terms—Motion planning, multi-query planning, link envelopes, reusable planning, configuration-space regions.

I. Introduction

Many high-degree-of-freedom manipulation workloads contain related queries in nearly fixed workcells. Robot kinematics persist, while start-goal pairs, local obstacles, or small scene edits change. Convex-region planners, including Iterative Regional Inflation (IRIS) and Graphs of Convex Sets (GCS), provide optimization-ready region-level abstractions [1]–[3]. Sampling-based planners avoid

region construction [4]–[7], but their reusable structures are typically non-volumetric roadmaps or query trees.

This article introduces RapidBoxForest (RBF), which uses reusable axis-aligned regions instead of only non-volumetric graphs while avoiding general convex-cover construction. It grows axis-aligned configuration-space boxes, checks them with link-envelope bounds, connects boxes that pass the configured scene-validation policy into a typed corridor graph, and reuses that graph across queries. This design makes the trade-off explicit: boxes are less expressive than general convex covers, but they are simpler to validate, connect, update, and reuse.

a) Problem statement: Let $\mathcal{Q} \subset \mathbb{R}^n$ be the joint-limit box of a fixed n -degree-of-freedom manipulator and let $\mathcal{O} \subset \mathbb{R}^3$ be the current workspace obstacle set. Given a set or stream of start-goal queries $(q_s, q_g) \in \mathcal{Q}^2$, where q_s and q_g are the start and goal configurations, RBF builds a finite family of configuration-space (C-space) boxes that pass the configured scene-validation policy. These policy-accepted boxes become the vertices of an adjacency graph. The design goal is to reduce scene-construction and maintenance overhead while retaining reusable volumetric planning structure and evidence whose validity is tied to explicitly conservative endpoint and link tests.

The geometric core adapts classical endpoint bounding. For rounded links, endpoint envelopes bound endpoint trajectories over a configuration-space box; their radius-expanded convex hull encloses the swept link and can certify C-space boxes when the endpoint source and separation tests are conservative. Conservative sources yield conditional box certificates, whereas tighter non-certifying sources remain filters whose paths require query validation. Reuse is split accordingly: scene-level RBF caches store one scene’s box forest and corridor graph, while the Lifelong Envelope Cache Tree (LECT) stores replayable kinematics-keyed endpoint and link evidence.

The evaluation mirrors this certificate model. Saved catalogs fix the start-goal and obstacle instances before planning, and success counts use post-run query validation with a 0.01 rad fixed segment step except for rows explicitly marked conservative. Shelf+KUKA LBR iiwa (Shelf+IIWA) rows with a prebuilt LECT cache include replay-based scene construction and query-time retrieval while excluding snapshot, memory, and setup costs. Update rows measure scene-cache maintenance, not post-edit query success or end-to-end replanning. The

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TABLE I

Reusable manipulation-planning paradigms. Abbrev.: PRM = probabilistic roadmap; RRT = rapidly-exploring random tree; BIT* = Batch Informed Trees; FMT* = Fast Marching Tree; IRIS = Iterative Regional Inflation; GCS = Graphs of Convex Sets; RBF = RapidBoxForest; LECT = Lifelong Envelope Cache Tree; LazyPRM = lazy PRM.

Family	Persistent object	Volumetric	Validation and certificate unit	Scene-edit reuse	Main trade-off
PRM and LazyPRM	Roadmap vertices and edges	No	Local edges or path segments	Topology can be reused; affected edges require rechecking	Reusable multi-query graph, but reusable evidence is non-volumetric.
RRT-Connect and RRT	Query tree	No	Incremental path segments	Little persistent reuse across queries	Query-focused discovery; persistent structure is query-local.
BIT* and FMT*	Batch sample graph or tree	No	Candidate graph edges	Primarily query-centric; limited scene-level reuse	Anytime quality improvement, but still edge-centric.
Adaptive cell decomposition	Configuration-space cells	Yes	Cells under exact, sampled, or interval tests	Explicit cells can be locally relabeled when maintained	Volumetric representation; scalability depends on the cell validator.
IRIS and GCS	Certified convex regions and graph	Yes	Convex collision-free regions	Region reuse is scene-coupled; edits may require reinflation or recertification	Optimization-ready regions with construction and recertification cost.
RBF	Box forest + typed graph + LECT evidence	Yes	C-space boxes via link envelopes; query validation outside certified regions	Scene boxes are maintained locally; LECT replays kinematics-keyed evidence	Volumetric reuse via axis-aligned boxes; certificates are conditional.

comparison characterizes reusable-region construction, not planner dominance or completeness.

This work makes five algorithmic contributions:

- 1) *Endpoint-to-link box validation*: a swept-capsule interval construction converts endpoint envelopes into link envelopes over a C-space box, yielding conditional box certificates when the endpoint source and separation test are conservative.
- 2) *Reusable box-forest planning*: RBF introduces a sample-guided construction that turns policy-accepted boxes into a reusable scene forest and typed corridor graph, separating certified box corridors from query-validated bridge, repair, and filter returns.
- 3) *Kinematics-keyed lifelong reuse*: LECT stores endpoint and link-envelope evidence under robot, descriptor, interval, and active-link keys, so later scene builds replay geometry bounds without importing stale obstacle labels or graph state.
- 4) *Edit-local scene maintenance*: obstacle deletions promote affected blocked domains after edited-scene validation, while insertions invalidate hit boxes and run bounded local refill instead of rebuilding the whole forest.
- 5) *Auditable multi-query evaluation*: saved catalogs, fixed-step post-run query validation, online timing definitions, and explicit cache accounting isolate reusable construction cost from query validation, simplification, snapshot, and memory effects.

The implementation is open source. Fig. 1 summarizes the method framework: link-interval computation above, RBF growth at lower left, and LECT reuse at lower right. Sections III–VI make these blocks precise and then evaluate them.

II. Related Work

Convex-region planners construct collision-free convex regions and search or optimize on the induced graph. IRIS variants inflate collision-free convex regions, and Graphs

of Convex Sets (GCS) make the resulting graph optimization explicit [1]–[3], [8]–[10]. RBF shares this region-level objective but restricts the regions to axis-aligned C-space boxes; the restriction limits expressivity while keeping validation, adjacency, and update operations local and explicit. Table I summarizes the qualitative position.

a) Continuous-collision-detection endpoint bounds: Swept-volume and interval-kinematics methods supply the geometric primitives used here. Convex-hull generator bounds and capsule radius lifts provide swept-set enclosures, while the Proximity Query Package (PQP), Gilbert–Johnson–Keerthi (GJK), the Flexible Collision Library (FCL), and interval Denavit–Hartenberg propagation provide collision-query and interval-kinematics primitives used in related continuous-collision-detection pipelines [11]–[22]. Redon *et al.* combine such primitives for articulated-body continuous collision detection over time steps [23], [24]; related Taylor-model and safety methods use similar swept-sphere bounds [25]–[27]. RBF uses configuration-space boxes rather than trajectory time steps as the bounding domain, stores compatible interval evidence in LECT, and assembles policy-accepted boxes into a typed corridor graph. Repaired, shortened, post-processed, or filter-only paths remain subject to query validation.

b) Graph reuse: Reusable graph planners amortize effort across related queries. Probabilistic roadmaps (PRMs) and LazyPRM reuse roadmaps and edge checks [6], [28]–[30], experience-based systems reuse prior paths [31], and adaptive cell methods split cells until they obtain a searchable accepted-cell graph [32], [33]. RBF combines reusable graph search with box-level envelope validation and a separate kinematics-keyed evidence cache.

c) Sampling and refinement: Rapidly-exploring random tree (RRT) variants, Batch Informed Trees (BIT*), Fast Marching Tree (FMT*), and the Open Motion Planning Library (OMPL) provide standard single-query planners, anytime references, and benchmarking infrastructure [7], [34]–[39]. Trajectory optimizers refine supplied

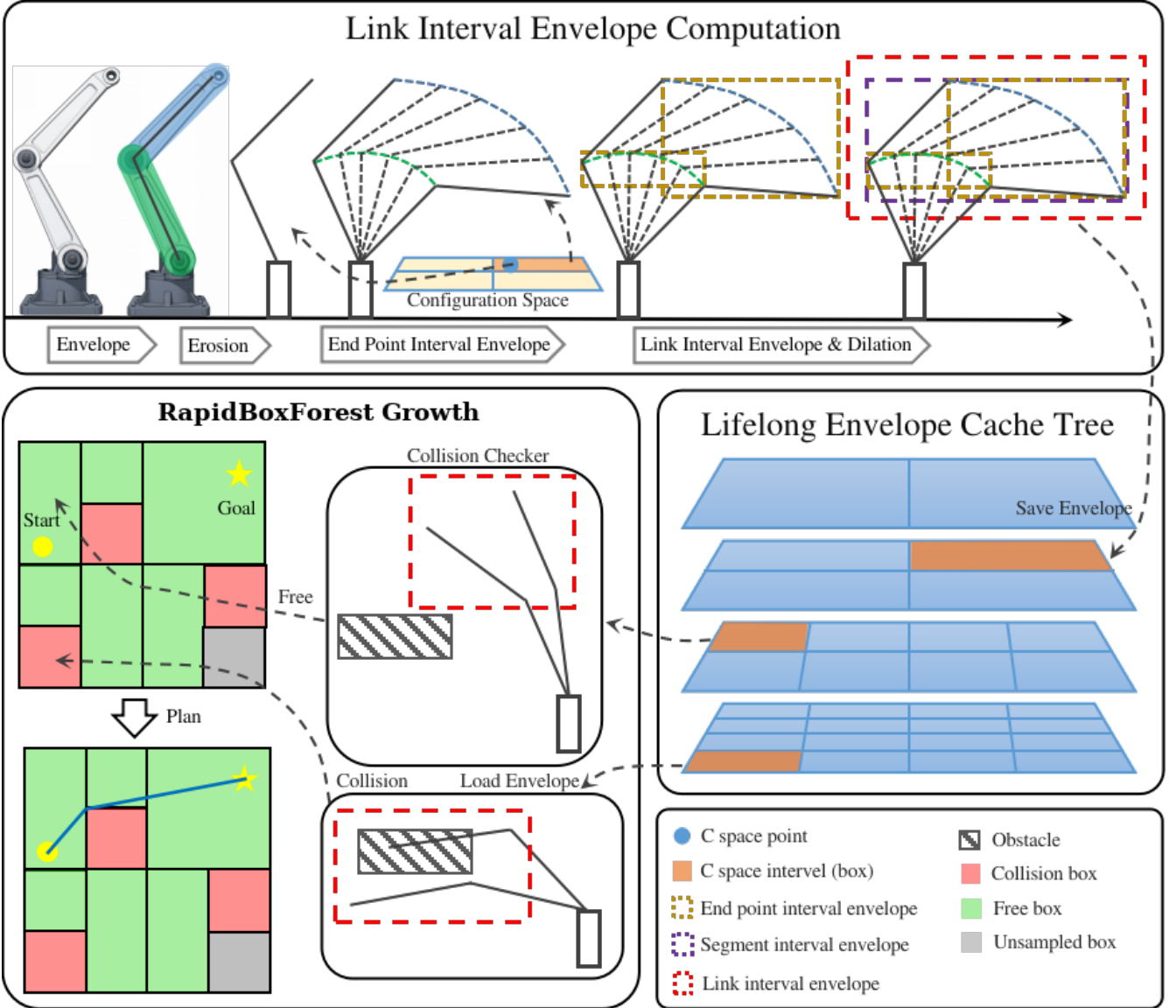


Fig. 1. RBF pipeline. Top: endpoint and segment interval envelopes induce link interval envelopes over a C-space interval. Lower left: sample-guided RBF growth labels free, collision, and unsampled boxes. Lower right: LECT stores kinematics-keyed envelope records for replay before scene filtering.

paths [40]–[43]. RBF supplies reusable volumetric context before repair, post-processing, and query validation.

III. Link Interval Envelopes

The link-envelope validation layer maps joint regions to endpoint and link bounds; graph connectivity and query acceptance are handled by later layers. For a joint region $I \subseteq \mathcal{Q}$, endpoint envelopes $E_k(I)$ bound endpoint trajectories for endpoint k , and link envelopes $B_\ell(I)$ bound rounded links for link ℓ over the same region:

$$I \mapsto \{E_k(I)\} \mapsto \{B_\ell(I)\}.$$

Axis-aligned C-space boxes are the default regions; local oriented bounding boxes (OBBs) in scaled joint coordinates are used only for typed bridge corridors. Endpoint evidence alone does not certify a box. It supports a certificate only after it is lifted to link envelopes and those envelopes pass conservative scene-separation tests.

Empirical or otherwise non-conservative endpoint sources are filters; candidate edges or paths that depend on them require query validation.

Assumption 1 (Collision model for certificates). *The certificate model covers only workspace-obstacle collision between active rounded links and $\mathcal{O}$; joint limits are enforced separately by requiring $I \subseteq \mathcal{Q}$. Self-collision, attachments, grippers, task regions, and other constraints are outside this certificate model unless encoded in the validation policy and cache descriptor. The experiments instantiate this subset with active-link versus workspace-obstacle collision checks.*

A. Geometric Foundation

Let C_0 be a body-frame shape. The rigid transform $T(q)x = R(q)x + t(q)$ maps a body-frame point x to workspace at configuration q , with rotation $R(q)$ and translation $t(q)$, and $\mathcal{S}_I(C_0)$ denotes the workspace set

swept by C_0 as q ranges over I . Let $\mathbb{B}_p(r)$ denote the radius- r ball in the p -norm, and let $\oplus$ denote Minkowski sum. For a capsule $C = S \oplus \mathbb{B}_2(r)$ with zero-radius center segment S , the fixed-radius ball commutes with rigid motion:

$$\mathcal{S}_I(C) = \mathcal{S}_I(S) \oplus \mathbb{B}_2(r).$$

Thus a rounded link can be bounded by first bounding the zero-radius center segment and then lifting by the link radius. For a convex body $P = \text{conv}(v_1, \dots, v_m)$ with vertices v_i , outer bounds on the vertex trajectories also bound the swept body,

$$\mathcal{S}_I(P) \subseteq \text{conv}\left(\bigcup_{i=1}^m \widehat{\Gamma}_i(I)\right).$$

Here $\widehat{\Gamma}_i(I)$ denotes a conservative outer bound on the workspace trajectory of vertex v_i , $i = 1, \dots, m$, over I . For a capsule link, only the two endpoint trajectories are needed. Let $\Gamma_a(I)$ and $\Gamma_b(I)$ denote the exact workspace trajectories of the proximal and distal endpoints over I . RBF uses these swept-sphere bounds over a configuration-space box domain and stores the resulting bounds as reusable kinematics-keyed bound records.

Proposition 1 (Endpoint envelopes induce link interval envelopes). *Consider a rounded link $L = [a, b] \oplus \mathbb{B}_2(r)$ and a joint interval I . If endpoint envelopes $E_a(I)$ and $E_b(I)$ satisfy $\Gamma_a(I) \subseteq E_a(I)$ and $\Gamma_b(I) \subseteq E_b(I)$, then*

$$\mathcal{S}_I(L) \subseteq \text{conv}(E_a(I) \cup E_b(I)) \oplus \mathbb{B}_2(r).$$

Consequently, any conservative outer representation of the right-hand side is a valid link interval envelope for the whole joint interval I .

Proof. For each $q \in I$, the center segment of the moved link is the convex hull of the two moved endpoints, and those endpoints lie in $\Gamma_a(I)$ and $\Gamma_b(I)$. Hence all center segments are contained in $\text{conv}(\Gamma_a(I) \cup \Gamma_b(I))$. The endpoint inclusions place this set inside $\text{conv}(E_a(I) \cup E_b(I))$. Adding the fixed radius ball commutes with the rigid-motion sweep of the capsule, giving the stated containment. $\square$

Corollary 1 (Conservative configuration-space box validation). *Let $I \subseteq \mathcal{Q}$ be an axis-aligned configuration-space box and let $\{B_\ell(I)\}$ be conservative link interval envelopes for all active rounded links. If $B_\ell(I) \cap \mathcal{O} = \emptyset$ for every link ℓ , then every configuration $q \in I$ is collision-free with respect to the collision model in Assumption 1.*

Proof. By Proposition 1, each conservative link envelope contains the swept workspace volume of its link over all configurations in I . This disjointness from $\mathcal{O}$ for every active link excludes workspace obstacle collision for the whole manipulator at every $q \in I$ under Assumption 1. $\square$

a) Numerical conservatism: A certificate is claimed only when endpoint propagation, radius padding, and obstacle separation produce outward bounds under the collision model of Assumption 1. For link radius r_ℓ , the

axis-aligned bounding box (AABB) broadphase pads by the infinity-norm ball $\mathbb{B}_\infty(r_\ell)$, which conservatively contains the Euclidean radius lift $\mathbb{B}_2(r_\ell)$. SupportHull/GJK denotes a support-function hull with GJK refinement; it is certificate-backed only with outward obstacle inflation, explicit margins, and conservative termination tolerances. Empirical endpoint sources and post-processing outside a conservative box union or a finite-cover OBB-corridor union are treated as non-conservative filters; they do not extend Corollary 1 or Theorem 1.

B. Endpoint Interval Envelopes

For endpoint k , let $\Gamma_k(I)$ be its exact trajectory over joint interval I . Any conservative set containing $\Gamma_k(I)$ is an endpoint interval envelope; the basic record is an endpoint axis-aligned bounding box (AABB) because it is compact and directly reusable by all link-envelope representations. Two certificate-backed endpoint sources are used. Interval forward kinematics (IFK) denotes the unsplit interval pass; IFK-AA is the affine-arithmetic implementation that propagates bounds along the Denavit–Hartenberg chain, with shared noise symbols preserving part of the trigonometric correlation. Hierarchical IFK (HIFK) bisects the joint interval and forms hulls of child endpoint boxes, reducing wrapping at higher materialization cost. HIFK is a hierarchical split-and-hull evaluation (for example, HIFK-5 means five split levels), so the split depth is part of the cache descriptor and cannot be exchanged with an unsplit record. A critical-sample source probes interval endpoints, midpoints, near-boundary configurations, and trigonometric turning candidates; it can reduce over-approximation relative to IFK, but does not certify coupled extrema. An analytical extrema source solves low-dimensional sinusoidal boundary subproblems, reducing one- and two-joint boundary cases to structured root search. It becomes conservative only with an interval-closure wrapper; the Analytical row in Table V is otherwise a comparison reference. Without such a wrapper, critical-sample and analytical endpoint sources are filters rather than certificate evidence. Any path using either source is accepted only after fixed-step query validation unless wrapped by an independent conservative certificate. During LECT tree descent, transient prefix transforms are reused so unaffected kinematic-chain prefixes are not recomputed. This affects runtime materialization only; it is not persistent scene state. The upper panel of Fig. 1 summarizes this endpoint-to-link conversion before the scene check accepts a box.

C. C-space OBB Endpoint Envelopes

Axis-aligned boxes remain the global forest primitive. Bridge and repair motions introduce local joint correlations. For a bridge transition to carry a certificate, RBF validates a local typed C-space OBB-corridor region with the finite-cover test below, while leaving the dyadic forest unchanged. Without a successful finite cover, the same transition remains only a typed candidate that requires

query validation. OBBs are defined in scaled joint coordinates. Let q_c be the local corridor center, and let $S \in \mathbb{R}^{n \times n}$ be a diagonal joint-scaling matrix with positive diagonal entries, defining the local metric. Let $m \leq n$ be the local dimension, $U \in \mathbb{R}^{n \times m}$ have scaled-metric orthonormal columns, and $h \in \mathbb{R}_{\geq 0}^m$ be the half-width vector. With the local cube $\Xi = \{\xi \in \mathbb{R}^m \mid -1 \leq \xi_i \leq 1, i = 1, \dots, m\}$ and $A = S^{-1}U \text{diag}(h) \in \mathbb{R}^{n \times m}$, the local region is

$$R_{\text{obb}} = \{q_c + A\xi \mid \xi \in \Xi\}.$$

The object R_{obb} is a configuration-space region, not a workspace OBB: the certificate comes from endpoint and link envelopes over R_{obb} , not from the fitted orientation.

a) *Cover certificate*: The conservative cover test tiles the local cube into subcubes X_τ , where τ indexes the cover tiles; for each $X_\tau \subseteq \Xi$, it forms the joint-space AABB hull

$$I_\tau = \text{bbox}_q(q_c + AX_\tau),$$

where $\text{bbox}_q(\cdot)$ denotes the joint-space AABB hull. If every I_τ passes the standard conservative link-envelope scene test, then

$$R_{\text{obb}} \subseteq \bigcup_{\tau} I_\tau \subseteq \mathcal{C}_{\text{free}},$$

where $\mathcal{C}_{\text{free}}$ is the collision-free configuration set under Assumption 1. When this test succeeds, the stored OBB certificate is a finite cover of axis-aligned box certificates.

b) *Correlated endpoint envelope*: A tighter source keeps the local OBB variables ξ correlated through the forward-kinematics computation. For each joint coordinate $j = 1, \dots, n$,

$$q_j = q_{c,j} + a_j^\top \xi, \quad \xi_i \in [-1, 1], i = 1, \dots, m,$$

where $a_j^\top$ is the j -th row of A . Affine-arithmetic propagation then produces endpoint records with center c_k , generators $g_{k,i}$, unit noise variables $\epsilon_i \in [-1, 1]$, and a coordinatewise nonnegative remainder vector ρ_k :

$$E_k(R_{\text{obb}}) = c_k + \sum_i g_{k,i} \epsilon_i \oplus [-\rho_k, \rho_k].$$

The sum runs over the affine-noise generators in the endpoint record. Equivalently, with $p_k(q_c)$ the endpoint position and $J_k(q_c)$ its Jacobian at the corridor center,

$$E_k(R_{\text{obb}}) = p_k(q_c) + J_k(q_c)A\Xi \oplus [-\rho_k, \rho_k],$$

where $\mu \in \{1, 2, 3\}$ indexes workspace coordinates, the sum is over local coordinates $a, b = 1, \dots, m$, and each coordinate remainder is bounded conservatively, for example by

$$\rho_{k,\mu} \geq \frac{1}{2} \sum_{a,b} |(A^\top H_{k,\mu} A)_{ab}|,$$

with $H_{k,\mu}$ an interval or otherwise conservative Hessian bound over R_{obb} . If these remainders are sampled rather than outward bounded, the OBB source is a non-conservative filter and not a certificate.

OBB endpoint records feed the same endpoint-to-link construction as axis-aligned C-space boxes. These records refine a local typed edge; they do not add OBB nodes to the global axis-aligned forest. For a workspace support

direction $d \in \mathbb{R}^3$, the support function of a zonotope-plus-box endpoint record is

$$h_{E_k}(d) = d^\top c_k + \sum_i |d^\top g_{k,i}| + \sum_{\mu=1}^3 |d_\mu| \rho_{k,\mu}.$$

Here the sum over i spans the affine-noise generators in the endpoint record, and μ indexes workspace coordinates. Because E_a and E_b enclose the endpoint trajectories, a conservative support bound for the zero-radius link centerline is

$$\bar{h}_\ell^\circ(d) = \max\{h_{E_a}(d), h_{E_b}(d)\},$$

and radius lifting gives the rounded-link support bound

$$\bar{h}_\ell(d) = \bar{h}_\ell^\circ(d) + r_\ell \|d\|_2.$$

Coordinate-axis queries of $\bar{h}_\ell$ provide an AABB broadphase, while the same support record feeds the SupportHull/GJK narrow phase. An OBB-corridor edge supports a certificate only when its endpoint bounds, finite-cover test, and separation tests are conservative; otherwise it is a typed transition that requires query validation. OBB-corridor records belong to the scene-local typed graph, not the persistent dyadic LECT tree.

D. Link-Envelope Representations

Endpoint records can be exported to multiple link representations without changing the upstream kinematic source. For a serial-chain capsule link ℓ whose proximal and distal endpoint records are $E_{\ell-1}(I)$ and $E_\ell(I)$, $B_\ell(I) = B_\ell^\circ(I) \oplus \mathcal{R}(r_\ell)$, with $B_\ell^\circ(I)$ bounding the centerline sweep and $\mathcal{R}(r_\ell)$ denoting the selected radius-lift set for link radius r_ℓ .

a) *AABB envelope*: AABB envelopes wrap the two endpoint envelopes in one box and pad by the link radius:

$$B_\ell(I) = \text{bbox}(E_{\ell-1}(I) \cup E_\ell(I)) \oplus \mathbb{B}_\infty(r_\ell).$$

Here $\text{bbox}(\cdot)$ denotes the workspace AABB hull. Segment subdivision may retain an explicit union of padded sub-envelopes for more selective filtering; collapsing that union back to one box would lose the tighter bound.

b) *Fixed-direction slab envelope*: A k -direction discrete oriented polytope (k -DOP) record stores lower and upper support values along a fixed direction set. Endpoint boxes are projected onto those directions, padded by the link radius, and merged over retained segment pieces. In the runtime pipeline, these slab tests form a low-cost intermediate filter: after the padded AABB broadphase, non-axis directions can reject close obstacle candidates before GJK. Slab tests are filters unless their endpoint source and separation tests are conservative.

c) *SupportHull envelope*: SupportHull keeps the two endpoint AABBs and link radius as a compact support record. For direction d , the support point is selected from the proximal or distal endpoint box, and obstacle refinement uses a fixed-size GJK narrow phase with outward obstacle inflation. The support record preserves more directional structure than a single AABB while keeping the cached record compact. When present, fixed-direction

slab records act only as an intermediate filter between the AABB broadphase and SupportHull/GJK refinement; they do not change the certificate condition, which still depends on conservative endpoint evidence and outward scene tests.

IV. Lifelong Envelope Cache Tree

LECT provides the kinematics-level cache beneath the scene-level RBF cache; it never caches scene-validity decisions. It stores scene-independent endpoint and link-envelope evidence keyed by robot kinematics and joint intervals. Obstacle tests, box labels, graph connectivity, and query outcomes remain scene-local, and every replayed envelope is checked against the current obstacle set.

A. Cache Key and Split Policy

LECT is a dynamically grown binary k -d tree over $\mathcal{Q}$. A node stores its joint interval, split metadata, cached endpoint and derived link-envelope evidence, and runtime residency state. The central invariant is that evidence is *kinematics-keyed* rather than *scene-keyed*: replay imports geometry bounds, not scene-validation labels or graph edges.

a) Evidence/label separation: An envelope record is replay-compatible only when the robot model, endpoint source, envelope representation, split descriptor, link radii, and active-link set match the current build. The split policy is part of this descriptor, so alternative policies define distinct cache descriptors. A descriptor match permits reuse of geometry bounds only; it never imports scene-validity labels. For a parent interval I , let I_d^- and I_d^+ be the two children obtained by splitting joint dimension d at its midpoint. Define the cumulative downstream envelope volume as

$$\mathcal{V}(I) := \sum_{\ell} \text{vol}(B_{\ell}(I)).$$

Here $\text{vol}(\cdot)$ denotes workspace volume. One implemented schedule chooses depth-wise split dimensions that minimize the worst-child cumulative envelope volume,

$$d^* = \arg \min_d \max \{ \mathcal{V}(I_d^-), \mathcal{V}(I_d^+) \}.$$

Depth-synchronous schedules keep replay deterministic; if the scheduled dimension is unavailable at a node, the runtime falls back to the widest valid dimension. Candidate dimensions are filtered by active-link reachability: splitting a joint that cannot affect any retained endpoint cannot tighten the downstream envelopes and is skipped for that descriptor.

b) Sparse interval staging: Planning cells need not materialize every ancestor. Each cell stores its exact joint interval, split counts, and descriptor fingerprint, which identifies the endpoint source, envelope representation, split policy, link radii, and active-link set used to interpret replay records. If no exact replay record is present, the runtime materializes the interval under that descriptor before validation.

B. Runtime Use and Validation Semantics

LECT serves local box materialization; graph search uses the scene forest. Given a joint interval, the runtime replays or materializes endpoint evidence, derives the requested link representation, and applies the current scene-validation policy. In the evaluated pipeline, AABB broadphase precedes SupportHull/GJK refinement. Parent or child records can accelerate materialization, but replayed evidence alone is not a collision-free certificate. When the required child records are present, their hulls can tighten an ancestor envelope; the reconstructed record remains representation-dependent and must still pass the current scene-validation policy. Directional records are staged filters: a joint interval first passes the padded AABB broadphase, and only the remaining close obstacle candidates reach slab or SupportHull/GJK refinement. If no exact replay record exists, the interval is materialized on demand and then handled by the same validation path, so a cache miss changes cost, not semantics or validation responsibility.

C. Storage and Warm-Build Semantics

Persistence is selective. Endpoint records and derived link-envelope records are the reusable evidence units. IFK+AABB records can be regenerated on demand. Non-AABB envelope records are stored only when their additional rejection power offsets lookup and storage cost. The prewarm pass fills target leaf envelope records and derives ancestor hulls from child evidence. This avoids recomputing direct forward kinematics independently at every ancestor depth. In a warm build, a cache hit reconstructs the requested link envelopes; the scene builder then checks those envelopes under the current scene-validation policy. Warm builds still choose and grow scene-specific boxes; no persisted box label is imported as a certificate. The Shelf replay used in the experiments is read-only during timing, so the replay store does not accumulate new evidence during those measurements. Random-scene cache growth is treated separately as part of the per-robot profile.

V. RapidBoxForest Construction

Fig. 2 illustrates the planning layer. RBF builds scene-specific policy-accepted C-space boxes and stores their adjacency in a typed corridor graph. For a fixed obstacle set, later queries use box membership and graph search instead of rebuilding the cover. LECT remains a separate kinematics-keyed evidence store, and each scene box is checked against the current obstacles.

A box is an axis-aligned joint interval $B = [l^{(1)}, u^{(1)}] \times \dots \times [l^{(n)}, u^{(n)}] \subseteq \mathcal{Q}$. A *policy-accepted box* has passed the configured scene-validation policy. Conservative link-envelope disjointness makes acceptance a collision-free certificate; a non-conservative filter makes the box only a planning region, so any path segment using it still needs query validation. Rejected cells are excluded from the forest, while selected blocked or inconclusive domains are retained for local refinement or later update.

Panel legends: end-effector (EE), configuration-space obstacle (C-obstacle), waypoint count (wpt), and θ_1 and θ_2 denote schematic joint coordinates.

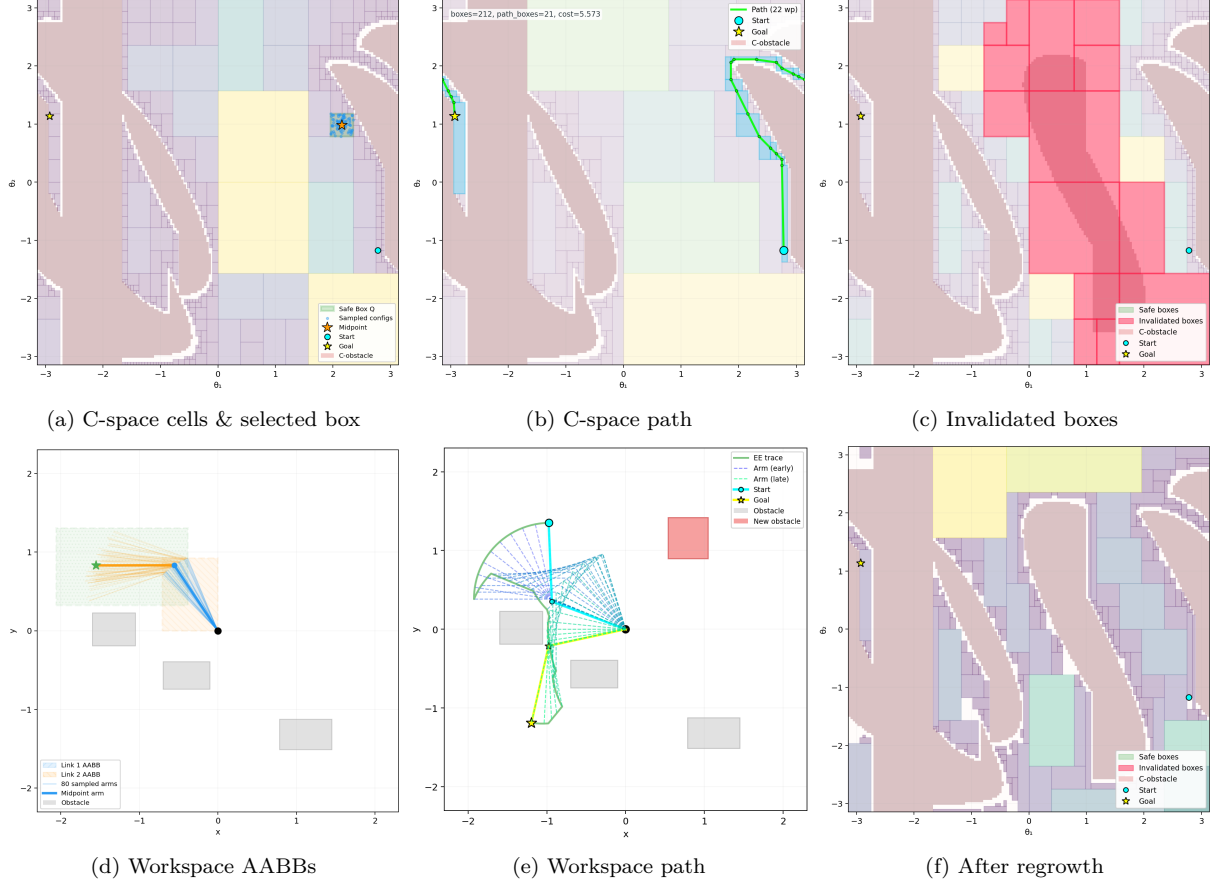


Fig. 2. Two-link schematic of the RBF build, query, and update cycle. A C-space box maps to workspace axis-aligned bounding boxes (AABBs); a box-corridor supplies the path; obstacle insertion triggers local refill. Panels show schematic logic, not experiment geometry; in (e), traces and arm labels denote schematic poses.

Algorithm 1 Local box materialization around a seed.

Input: Seed configuration q , LECT tree $\mathcal{T}_L$, obstacle set $\mathcal{O}$, depth cap $D_{\max}$
Output: Policy-accepted interval or failure
1: start from the root interval containing q with depth 0
2: **while** depth $\leq D_{\max}$ **do**
3: replay or materialize current endpoint and link evidence
4: **if** the current interval passes the scene-validation policy **then**
5: **return** current interval
6: **end if**
7: split, increment depth, and descend to the child containing q
8: **end while**
9: **return** failure

A. Construction Objective

Given the joint-limit box $\mathcal{Q}$ and obstacle set $\mathcal{O}$, let $V_{\mathcal{O}}(B)$ denote the configured scene-validation predicate for a box B . The build stage constructs a forest $\mathcal{F}$ of N retained boxes satisfying

$$\mathcal{F} = \{B_1, \dots, B_N\},$$

$$B_i \subseteq \mathcal{Q}, \quad V_{\mathcal{O}}(B_i), \quad i = 1, \dots, N.$$

The reusable output is $(\mathcal{F}, G)$, where G is the typed corridor graph storing adjacency and auxiliary transitions; the box budget targets anchors, frontiers, and connectors.

For two boxes

$$B_i = \prod_{d=1}^n [l_i^{(d)}, u_i^{(d)}], \quad B_j = \prod_{d=1}^n [l_j^{(d)}, u_j^{(d)}],$$

let $\varepsilon_c \geq 0$ denote the connection tolerance. RBF treats the boxes as connected when the closed intervals in every coordinate overlap or have a gap no larger than ε_c :

$$\max\{l_i^{(d)}, l_j^{(d)}\} \leq \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \quad d = 1, \dots, n.$$

Only exact overlap or contact directly yields a contained box-corridor. Tolerance-gap and explicit segment edges are graph-search aids, not certificates, unless they are converted into conservative box-corridor or finite-cover OBB-corridor regions; otherwise they require query validation.

B. Build Pipeline

The build proceeds through four stages. First, local materialization descends the LECT tree around a seed and returns the first interval accepted along that descent. Second, a selected leaf-cell sweep classifies cells and records blocked domains for refinement or edits. Third, frontier growth adds boxes near anchors and query-relevant boundaries. Finally, consolidation merges or removes redundant boxes only when the resulting boxes still satisfy the same validation rule. Algorithms 1 and 2 define the two validation routines reused throughout this pipeline; frontier growth and consolidation repeat the same materialize-validate-connect pattern under budget limits and do not introduce a separate validation primitive.

Algorithm 2 Adaptive leaf sweep (selected leaf-cell pass) and local refinement.

Input: LECT tree $\mathcal{T}_L$, obstacle set $\mathcal{O}$, sweep budget

Output: Initial forest $\mathcal{F}$ and retained collision domains $\mathcal{C}$

- 1: classify priority cells by replaying or materializing LECT evidence
 - 2: insert cells that pass validation into $\mathcal{F}$; prune covered descendants
 - 3: retain inconclusive or blocked cells in $\mathcal{C}$
 - 4: refine only selected domains in $\mathcal{C}$ using local materialization
 - 5: insert refined boxes that pass validation, remain inside their domain, and connect to the forest
 - 6: **return** $\mathcal{F}, \mathcal{C}$
-

a) Controls: Validation semantics and timings are governed by the build budget, LECT descriptor, endpoint and link representation, admitted edge types, and validation resolution. Materialization failures remain deferred domains for restricted refinement; frontier and repair witnesses carry certificates only after conversion into conservative box-corridor or finite-cover OBB-corridor regions. Lower-level scheduling constants are fixed in the experiment configuration. In Algorithms 1 and 2, $\mathcal{T}_L$, $D_{\max}$, and $\mathcal{C}$ denote the LECT tree, the descent depth cap, and the retained blocked or inconclusive collision-domain cache.

C. Corridor Query and Certificate

After consolidation, G has policy-accepted vertices and edge types $E_{\cap}$ (overlap/contact), E_{ε} (tolerance gap), E_s (segment), E_o (finite-cover OBB-corridor), and E_{π} (repair witness). Overlap/contact edges and finite-cover OBB-corridor edges carry certificates only when backed by conservative region checks. Repair-witness edges carry certificates only after conversion into a conservative box-corridor or OBB-corridor representation. Tolerance-gap, segment, unconverted repair-witness, and other outputs require query validation. Certificate status is determined by the backing conservative region check, not by the edge label alone.

Given start and goal configurations, the query locates containing boxes or configured anchor boxes and searches G . The certificate applies only to returned paths contained in conservative box regions or finite-cover OBB-corridor regions. Theorem 1 states the box-region case; Section III-C gives the finite-cover certificate for OBB-corridor edges. All other returns are query-validated results and carry no certificate claim.

Theorem 1 (Conditional conservative box-corridor path). *Let $\mathcal{F}$ be a forest whose boxes have each passed conservative link-envelope validation against obstacle set $\mathcal{O}$. Let $R_{\mathcal{F}} := \bigcup_{B \in \mathcal{F}} B$ denote the represented box union. If a continuous path $\gamma : [0, 1] \rightarrow \mathcal{Q}$ satisfies $\gamma([0, 1]) \subseteq R_{\mathcal{F}}$, then every configuration on γ is collision-free with respect to the collision model in Assumption 1.*

Proof. For any $s \in [0, 1]$, containment gives a conservatively validated box $B \in \mathcal{F}$ with $\gamma(s) \in B$. By conservative link-envelope validation and Corollary 1, the robot geometry at $\gamma(s)$ is disjoint from $\mathcal{O}$. Since s was arbitrary, the whole path is collision-free. $\square$

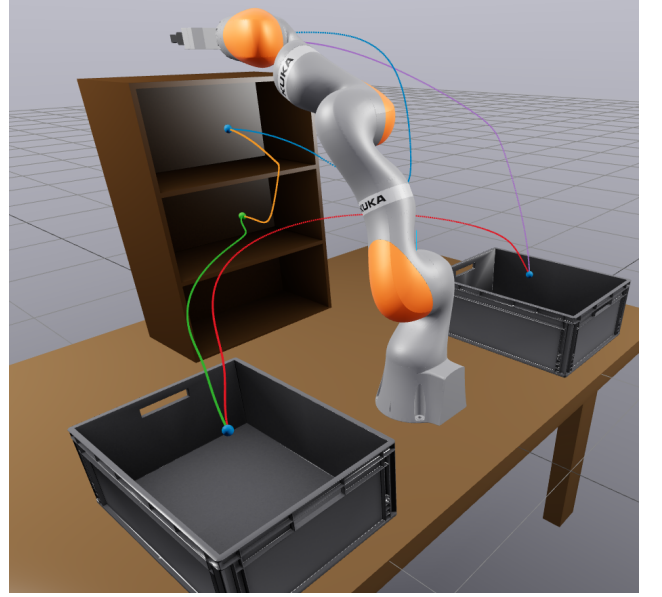


Fig. 3. Canonical Marcucci Shelf+KUKA LBR iiwa (Shelf+IIWA) scene with saved anchors: approach-side (AS), target-shelf (TS), center-shelf (CS), left-bin (LB), and right-bin (RB).

Certificate interpretation. The theorem certifies collision-freedom for paths contained in conservatively validated regions. It does not imply coverage or completeness: finite depth, axis-aligned boxes, envelope overapproximation, and bridge or repair limits can miss paths. Paths that use uncovered space, include residual non-corridor segments, are post-processed, or traverse boxes admitted by non-conservative filters are accepted only if independent query validation succeeds.

VI. Experiments

The evaluation combines two planning settings with two mechanism studies. The Shelf+KUKA LBR iiwa (Shelf+IIWA) profile fixes the RBF configuration before summarization. The random-scene catalogs store all queries and obstacles before planner execution. Across planning rows, success, validation, online timing, simplification, and path-length-ratio rules are fixed. Offline curves sweep planner budget explicitly.

a) Protocol: Timings were measured on Ubuntu 22.04 with an Intel i5-12600KF CPU and 31 GiB RAM. For reproducibility, each generated table or figure stores data sources, thread policies, simplification settings, checksums, and integrity metadata. The RBF and LECT computations use an 8-thread budget, with numerical library thread limits set to the same budget. The OMPL implementations of RRT-Connect, PRM, and BIT* are single-threaded per-query invocations under this protocol; the timings therefore document the stated thread policy, not equal-core scaling. In the planning tables, Build is reusable scene-construction time before the timed queries. Online/q is per-query planner time after a reusable build, excluding final simplification and query validation after the run; in cross-algorithm timing columns, $T = \text{Online}/q$, not reusable build time.

Success requires post-run query validation with the fixed 0.01 rad segment step. Planning table entries report median $[Q_1, Q_3]$, where Q_1 and Q_3 are the first and third quartiles, unless a caption or note states otherwise. Amortized entries of the form Build/ K + Online/ q use K as the stated reuse horizon, not as a success count or validation step. Each validated path-length ratio is the returned path length after fixed-step validation divided by a same-query empirical reference computed only from successful validated runs at the same operating point. Unless a table note states otherwise, the same 0.01 rad validation step is used for the reference and returned path. The symbols L^* and L_q^* denote empirical reference lengths: L^* is the saved Shelf reference and L_q^* is the per-query reference used in random-scene rows; the star marks a reference length, not an optimality certificate. Smaller ratios indicate shorter validated paths relative to those references. Mechanism-study tables follow the aggregation stated in their notes. The main planning tables summarize query-validated outcomes rather than corridor certificates. Planning rows use the collision model in Assumption 1: active rounded links are checked against $\mathcal{O}$. Self-collision, attached objects, grippers, and task-specific forbidden regions require additional scene checks before a certificate is claimed.

b) *Cache and profile accounting*: Bridge and repair witnesses become certificate-backed after conservative box-corridor or finite-cover OBB-corridor conversion; otherwise they remain query-validated transitions. Warm-cache rows include compatible LECT lookup and evidence reads; they exclude snapshot creation/loading, resident memory, and offline setup amortization. The d23 label identifies a depth-23 LECT cache: 16,777,215 records, 12.75 GiB on disk, and 958.079 s prewarm time.

A. Shelf+IIWA Fixed-Profile Study

Fig. 3 shows the Shelf+IIWA scene and fixed five-anchor query cycle with approach-side (AS), target-shelf (TS), center-shelf (CS), left-bin (LB), and right-bin (RB) anchors; the executed order is AS-TS-CS-LB-RB-AS. Table II and Fig. 4 report the fixed 100-box Shelf profile (b100) over seeds 0–7. The cycle is used for mechanism ablations and profile selection; saved random-scene catalogs below test broader start-goal variation.

The b100 RBF baseline uses SupportHull link envelopes replayed from the d23 cache: Build 0.171 s, Online/ q 0.077 s, and five-query amortized time 0.111 s; Table II gives quartiles. The Live HIFK-5 row keeps the same depth and validation budget but computes evidence live, giving a 2.770 s reusable build; the other rows replay d23 evidence. After OBB-corridor conversion, the 40 validated Shelf paths in Table II contain no residual non-corridor segments. The representation rows isolate implementation choices: Link AABB changes Online/ q from 0.077 s to 0.083 s while changing the median L/L^* from 1.15 to 1.19; SupportHull-only raises the five-query amortized time from 0.111 s to 0.126 s.

The Shelf profile table reports query-validated pipeline behavior; only converted corridor segments carry certificates. In that fixed pipeline, all 40 paths pass validation at the 0.01 rad step. With the query bridge, repair, short-cutting, and final simplification all disabled, the ablated pipeline yields 0/40 paths; this is a full-pipeline consistency check, not a single-stage failure diagnosis.

B. Shelf Cross-Algorithm Context

The cross-algorithm shelf study applies the same five queries, 0.01 rad validation segment step, and 0.010 s final-simplification budget to RBF, IRIS/GCS, PRM, RRT-Connect, and BIT*. RBF uses the b100 profile and per-query Online/ q ; PRM and IRIS/GCS report reusable builds, while RRT-Connect and BIT* are single-query rows with no reusable build term. BIT* checkpoints are offline quality–time references, not online stopping rules.

Under these definitions, reusable builds are 0.171 s (RBF), 20.040 s (PRM), and 355.425 s (IRIS/GCS). The RBF build excludes d23 prewarm, snapshot loading costs, and resident memory. In the per-query T columns, RRT-Connect has lower T than RBF in all five Shelf queries but has higher L/L^* in all five; BIT* has higher T than RBF but lower L/L^* in all five. This timing–quality comparison is limited to the saved five-query cycle and its stated reusable build accounting.

C. Random Multi-Robot Scenes

The random-scene study uses saved random-scene catalogs: each scene has ten pre-generated queries shared by every method. Across KUKA LBR iiwa, Universal Robots UR5, and Franka Emika Panda, the medium and hard catalogs over ten seeds contain 600 query instances before applying the method-independent pruning rule described next. For each robot and seed, medium and hard rows share the saved start-goal set; hard rows are formed by appending a saved obstacle-prefix extension to the medium row, so difficulty changes are nested rather than regenerated. Before summarizing any method, a fixed, method-independent pruning rule removes cases where the direct start-goal segment already validates without planning. The pruning changes summary statistics but not the saved catalogs, which retain ten queries per scene. The trade-off plot uses the five-query amortized quantity Build/5 + Online/ q for rows with reusable builds and Online/ q otherwise as a fixed display point; K is a reuse horizon, not a post-pruning retained-query count. Table IV and Fig. 6 summarize the operating points and corresponding checkpoint curves. RBF builds one query-agnostic forest per scene and reuses it for ten queries; RBF random rows use LECT caches and anchors tied to each robot rather than Shelf-specific assets. Each RBF row in Table IV uses the OBB-corridor profile and passes fixed-step query validation on the summarized query set. In those RBF rows, accepted paths use only converted finite-cover C-space OBB-corridor repair witnesses; no residual non-corridor segments remain. PRM is built once per scene, whereas

TABLE II
Shelf+IIWA RBF profile and mechanism ablations.

Case	Build	Online/q	Amort. 5q	L/L^*
RBF baseline	0.171 [0.166, 0.174]	0.077 [0.072, 0.085]	0.111 [0.107, 0.117]	1.15 [1.08, 1.39]
Critical sample	0.279 [0.275, 0.283]	0.087 [0.078, 0.095]	0.145 [0.134, 0.151]	1.14 [1.09, 1.38]
Link AABB	0.167 [0.164, 0.171]	0.083 [0.077, 0.086]	0.116 [0.111, 0.119]	1.19 [1.08, 1.48]
Live HIFK-5	2.770 [2.644, 3.003]	0.181 [0.139, 0.207]	0.739 [0.703, 0.777]	1.15 [1.08, 1.43]
SupportHull only	0.193 [0.188, 0.200]	0.087 [0.082, 0.095]	0.126 [0.120, 0.136]	1.15 [1.08, 1.39]

Notes: Median $[Q_1, Q_3]$, s. Build is reusable; Amort. 5q = Build/5+Online/q; Online/q excludes simplification/validation. AABB = axis-aligned bounding box; HIFK = hierarchical interval forward kinematics; SupportHull = support-function hull. Critical sample is non-certifying; L/L^* uses the same-query 0.01 rad reference.

Labels: IRIS/GCS = Iterative Regional Inflation + Graphs of Convex Sets; RBF b100 = fixed 100-box Shelf point; Amort. 5q = Build/5 + Online/q.

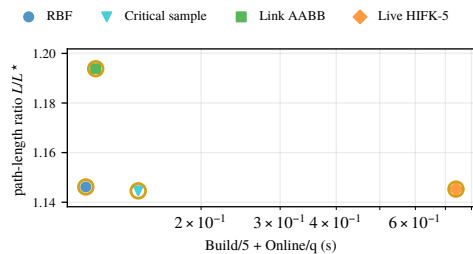


Fig. 4. Shelf+IIWA mechanism trade-off. Points plot Table II Amort. 5q versus L/L^* ; smaller ratios are shorter, and gold rings mark Table II rows.

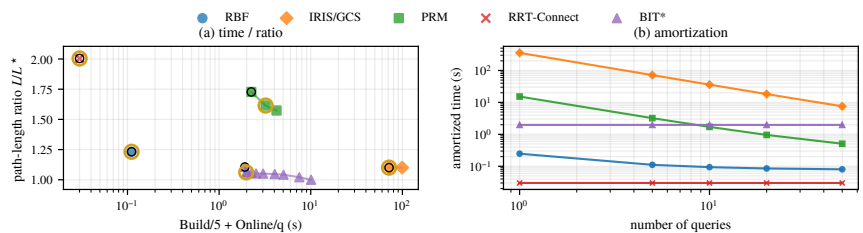


Fig. 5. Shelf+IIWA cross-algorithm trade-off. Black rings mark first successful checkpoints. The horizontal coordinate is Build/5 + Online/q for reusable methods and Online/q for single-query rows; smaller L/L^* is shorter. Right: amortized time per query versus reuse count.

TABLE III
Shelf+IIWA query-validated cross-algorithm comparison.

Query	RBF b100 Build 0.171 s		IRIS/GCS Build 355.425 s		PRM Build 20.040 s		RRT-Connect No reusable build		BIT* No reusable build	
	T (s)	L/L^*	T (s)	L/L^*	T (s)	L/L^*	T (s)	L/L^*	T (s)	L/L^*
AS→TS	0.065 [0.064, 0.070]	1.40 [1.34, 1.65]	0.449 [0.449, 0.449]	1.44 [1.44, 1.44]	0.198 [0.147, 0.248]	1.61 [1.42, 1.73]	0.034 [0.024, 0.052]	2.27 [2.05, 2.72]	4.004 [4.002, 4.004]	1.06 [1.02, 1.08]
TS→CS	0.085 [0.082, 0.089]	1.13 [1.11, 1.14]	0.269 [0.269, 0.269]	1.26 [1.26, 1.26]	0.173 [0.118, 0.312]	2.99 [2.11, 3.22]	0.054 [0.021, 0.089]	3.98 [2.80, 5.09]	10.138 [10.004, 10.386]	1.09 [1.04, 1.11]
CS→LB	0.073 [0.070, 0.077]	1.47 [1.32, 1.82]	0.381 [0.381, 0.381]	1.07 [1.07, 1.07]	0.136 [0.120, 0.195]	1.49 [1.39, 2.09]	0.032 [0.027, 0.070]	2.20 [1.75, 2.51]	7.504 [7.504, 7.725]	1.03 [1.01, 1.06]
LB→RB	0.078 [0.075, 0.083]	1.09 [1.07, 1.22]	0.786 [0.786, 0.786]	1.07 [1.07, 1.07]	0.164 [0.091, 0.399]	1.18 [1.09, 1.27]	0.001 [0.001, 0.002]	1.29 [1.27, 1.32]	1.654 [1.654, 1.654]	1.02 [1.02, 1.03]
RB→AS	0.081 [0.077, 0.086]	1.04 [1.03, 1.05]	0.527 [0.527, 0.527]	1.01 [1.01, 1.01]	0.252 [0.096, 0.295]	1.23 [1.17, 1.24]	0.001 [0.001, 0.001]	1.07 [1.03, 1.10]	0.204 [0.204, 0.204]	1.02 [1.01, 1.03]

Notes: Median $[Q_1, Q_3]$, s. Header Build is reusable; T = Online/q excludes simplification/validation. L/L^* uses the same-query 0.01 rad reference. IRIS/GCS = Iterative Regional Inflation/Graphs of Convex Sets; PRM = probabilistic roadmap; RRT-Connect = bidirectional RRT; BIT* = Batch Informed Trees. Anchors as in Fig. 3.

RRT-Connect and BIT* solve each query independently. The random-scene comparison omits an IRIS/GCS row: the fixed saved-catalog rule requires full validated success before computing comparable Online/q and path-length-ratio statistics. Thus Table IV and Fig. 6 compare RBF, PRM, RRT-Connect, and BIT* under shared catalogs. RRT-Connect has the lowest median Online/q in five of the six operating points; in Panda-Hard, RBF has the lowest median Online/q. After fixed-step validation, the RRT-Connect median validated path-length ratio exceeds the RBF ratio in all six points. With scene-forest reuse, RBF has lower median Online/q than BIT* in all six points; L/L_q^* matches BIT* to table precision in the two IIWA rows and is higher than BIT* in the four UR5/Panda rows.

D. Mechanism Studies

Tables V and VI compare endpoint sources and envelope narrow phases. Their numeric width columns are joint-interval side lengths in radians. Table V reports endpoint volume as a ratio to the Analytical row at the same width and the sampled-union signed gap in millimeters. Negative gaps indicate sampled undercoverage relative to the finite sampled union, not certificate margins. Only IFK-AA and HIFK rows are certificate-backed; Critical sample, Analytical, and Monte Carlo rows are comparison or reference rows rather than certificate evidence. Table VI motivates the AABB broadphase: SupportHull/GJK stores a compact support record, but wide boxes increase collision-test time when many obstacle candidates reach the GJK narrow phase. These rows are mechanism studies rather

TABLE IV
Saved-catalog random-scene operating points.

Scenario	RBF			PRM			RRT-Connect		BIT*	
	Build	Online/q	L/L_q^*	Build	Online/q	L/L_q^*	Online/q	L/L_q^*	Online/q	L/L_q^*
IIWA-Medium	0.075 [0.058, 0.083]	0.009 [0.007, 0.011]	1.00 [1.00, 1.03]	3.567 [3.007, 3.933]	0.038 [0.014, 0.070]	1.03 [1.02, 1.12]	0.002 [0.001, 0.002]	1.02 [1.01, 1.05]	0.107 [0.087, 0.154]	1.00 [1.00, 1.01]
IIWA-Hard	0.097 [0.086, 0.105]	0.014 [0.011, 0.015]	1.00 [1.00, 1.03]	4.175 [3.408, 5.637]	0.048 [0.021, 0.102]	1.03 [1.02, 1.08]	0.002 [0.002, 0.003]	1.03 [1.00, 1.07]	0.129 [0.097, 0.186]	1.00 [1.00, 1.01]
UR5-Medium	0.262 [0.254, 0.270]	0.075 [0.020, 0.086]	1.05 [1.00, 1.14]	3.124 [2.966, 3.418]	0.323 [0.085, 0.426]	1.02 [1.00, 1.12]	0.023 [0.002, 0.049]	1.23 [1.06, 1.54]	0.134 [0.101, 0.167]	1.01 [1.00, 1.08]
UR5-Hard	0.276 [0.264, 0.284]	0.083 [0.043, 0.108]	1.05 [1.01, 1.14]	2.488 [2.325, 3.048]	0.419 [0.129, 0.437]	1.06 [1.01, 1.19]	0.048 [0.008, 0.090]	1.36 [1.14, 1.58]	0.154 [0.115, 0.171]	1.03 [1.00, 1.13]
Panda-Medium	0.250 [0.218, 0.265]	0.028 [0.013, 0.039]	1.05 [1.02, 1.09]	3.431 [3.364, 4.180]	0.066 [0.038, 0.159]	1.06 [1.03, 1.08]	0.005 [0.002, 0.011]	1.22 [1.12, 1.35]	0.116 [0.092, 0.152]	1.00 [1.00, 1.00]
Panda-Hard	0.224 [0.221, 0.237]	0.043 [0.021, 0.053]	1.05 [1.01, 1.16]	2.337 [1.627, 3.978]	0.054 [0.034, 0.126]	1.10 [1.05, 1.16]	0.056 [0.019, 0.136]	1.25 [1.14, 1.37]	0.164 [0.123, 0.214]	1.01 [1.00, 1.09]

Notes: Median $[Q_1, Q_3]$, s. IIWA/UR5/Panda = KUKA LBR iiwa/Universal Robots UR5/Franka Emika Panda. Direct successes are pruned; catalogs retain ten queries/scene. Build is reusable for RBF/PRM; RRT-Connect/BIT* are single-query. Online/q excludes simplification/validation; L/L_q^* uses the same-query 0.01 rad reference.

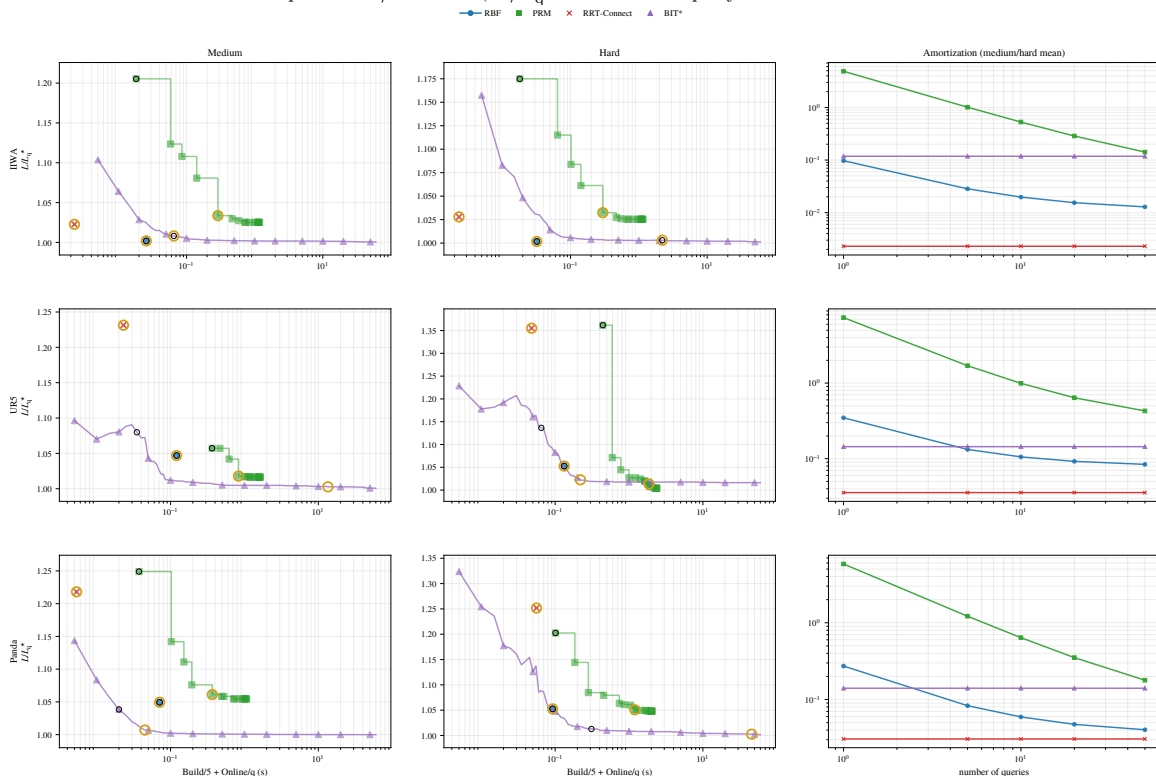


Fig. 6. Saved-catalog random-scene trade-off. Left/middle: fixed five-query point Build/5 + Online/q; right: reuse horizon K in Build/ K + Online/q. Online/q is per-query online time; smaller path-length ratios are shorter. Rows are IIWA, UR5, and Panda. No-build methods plot at Online/q. Rings mark first validated checkpoints (black) and later $\geq 1\%$ path-ratio gains (gold); PRM and BIT* use offline checkpoints.

than end-to-end planning rows. They explain the default implementation choices used by the planning experiments. In Table V, HIFK-5 keeps endpoint volumes much closer to the Analytical reference than IFK-AA at large widths while retaining certificate-backed interval construction. The Critical sample and Monte Carlo rows expose sampled-union gaps and therefore remain references, not sources of certificates. In Table VI, Link AABB has larger broadphase volume than SupportHull/GJK at narrow widths, but its build cost is lower or comparable and its hot-loop collision-test cost is substantially lower. The planner therefore uses AABBs as the broadphase representation and leaves exact collision checks to the final query-validation path. This separation keeps the paper claims aligned with the implementation: certificate rows

explain conservative coverage, while sampled and support-function rows explain cost or reference behavior only.

E. Dynamic Updates

Table VII reports scene-cache maintenance only: its warm-build rows are scene-cache rebuilds, and no query bridge, connector, simplification, audit, or query timing is included.

A diagnostic audit on locally updated forests and matching freshly rebuilt forests, each without the query bridge and repair stages, produced 0/8 paths in both diagnostic configurations. The 0/8 result is a scope check: without bridge and repair, the maintenance-only configuration is not an end-to-end planning pipeline, so the row should not be read as post-edit query success or a replanning benchmark.

TABLE V
Endpoint-envelope sources over joint-interval side lengths.

Source	Volume ratio					Time (μ s)					Gap (mm)				
	0.02	0.10	0.20	0.30	0.50	0.02	0.10	0.20	0.30	0.50	0.02	0.10	0.20	0.30	0.50
IFK-AA	1.09	1.52	2.30	3.53	8.61	5.48	6.45	6.58	6.81	7.35	—	—	—	—	—
HIFK-3	1.04	1.22	1.51	1.89	3.12	25.94	29.16	27.21	29.27	29.38	—	—	—	—	—
HIFK-5	1.02	1.13	1.30	1.51	2.07	70.40	73.82	73.15	75.92	72.32	—	—	—	—	—
Critical sample	1.00	1.00	1.00	0.99	0.98	8.68	10.03	11.56	12.69	16.33	-0.05	-1.17	-4.53	-10.77	-30.97
Analytical	1.00	1.00	1.00	1.00	1.00	6429.96	6602.25	6800.10	6927.12	7096.01	0.00	0.00	0.00	0.00	-4.49
Monte Carlo	0.64	0.74	0.78	0.79	0.82	2251.74	11233.26	22352.95	33813.58	56835.57	-6.05	-32.68	-65.57	-88.87	-123.53

Notes: Widths are rad. Volume is normalized to Analytical; time is endpoint construction. IFK-AA = affine-arithmetic interval forward kinematics; HIFK = hierarchical interval forward kinematics. IFK-AA/HIFK certify; other rows are references. Gap = sampled-union signed gap; — = omitted.

TABLE VI
Link-envelope representations over joint-interval side lengths.

Envelope	Volume (m^3)						Build (μ s)						Test (μ s)					
	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50
Link AABB	0.0136	0.0201	0.0384	0.147	0.508	4.61	0.138	0.152	0.167	0.166	0.171	0.178	0.586	0.605	0.601	0.604	0.630	0.790
SupportHull	0.000359	0.00266	0.0143	0.106	0.449	4.54	0.205	0.214	0.218	0.219	0.217	0.250	12.520	12.850	12.267	12.223	12.659	13.026

Notes: Widths are rad; volume is mean workspace volume. AABB = axis-aligned bounding box; SupportHull = support-function hull. Test pools far/local probes; build excludes endpoint envelopes.

TABLE VII
Adaptive leaf-sweep maintenance-only timing.

Operation	Time (ms)	Speedup
Warm build (10 obstacles)	922.75 [897.57, 955.67]	—
Insert to 15 obstacles	6.06 [5.69, 6.54]	153.3–207.5 $\times$
Remove from 15 obstacles	0.35 [0.31, 1.65]	943.6–3354.0 $\times$
Warm build (15 obstacles)	958.79 [941.67, 1115.18]	—

Notes: Median $[Q_1, Q_3]$, ms. Maintenance only; speedup is relative to the 15-obstacle warm build.

VII. Conclusion

In summary, RBF validates axis-aligned boxes with endpoint-to-link swept-capsule bounds, connects accepted boxes in a scene-level forest and typed corridor graph, and stores reusable kinematics evidence in LECT. Certificate claims apply only to conservative box unions or finite-cover OBB-corridor unions; all other paths are accepted only after query validation.

Saved-catalog Online/q excludes final simplification and post-run validation. For Shelf+IIWA, RBF replay from the depth-23 LECT cache gives median 0.171 s reusable build, 0.077 s Online/q, and 0.111 s five-query amortized time; live HIFK-5 under the same depth and validation budget takes 2.770 s. In saved random-scene catalogs, RRT-Connect has lower median Online/q in five of six points, but its median validated path-length ratio exceeds the corresponding RBF ratio in all six points. Compared with BIT*, RBF has lower median Online/q in all six points; BIT* returns lower validated path-length ratios in the four UR5/Panda operating points. Non-corridor or filter-derived returns remain query-validated, not certified, unless converted into conservative box-corridor or finite-cover OBB-corridor evidence. Snapshot and memory costs are outside the timing protocol, and non-obstacle validators require explicit validation-policy encoding before they are included in certificate claims.

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